

38. .Develop a domain-specific chatbot using LangChain and a vector database.

Code:

```
from sentence_transformers import SentenceTransformer
from transformers import pipeline
import faiss
import numpy as np

# =====
# DOMAIN-SPECIFIC CHATBOT
# =====

print("=====")
print("    AI DOMAIN-SPECIFIC CHATBOT")
print("=====")

# =====
# 1. DOMAIN DOCUMENTS
# =====

documents = [
    "Artificial Intelligence is a branch of computer science "
    "that creates intelligent systems.",

    "Machine Learning is a subset of Artificial Intelligence "
    "that allows computers to learn from data.",

    "Deep Learning is a type of Machine Learning that uses "
```

"neural networks with multiple layers.",

"Natural Language Processing helps computers understand "

"and process human language.",

"Computer Vision enables computers to understand images "

"and videos.",

"Reinforcement Learning allows an agent to learn by "

"receiving rewards and penalties."

]

```
print("\nDocuments loaded:", len(documents))
```

```
# =====
```

```
# 2. LOAD EMBEDDING MODEL
```

```
# =====
```

```
print("\nLoading embedding model...")
```

```
embedding_model = SentenceTransformer(
```

```
    "all-MiniLM-L6-v2"
```

```
)
```

```
print("Embedding model loaded.")
```

```
# =====
```

```
# 3. CREATE DOCUMENT EMBEDDINGS
```

```
# =====
```

```
print("\nCreating document embeddings...")
```

```
document_embeddings = embedding_model.encode(  
    documents  
)
```

```
document_embeddings = np.array(  
    document_embeddings  
)  
.astype("float32")
```

```
print("Embeddings created.")
```

```
# =====
```

```
# 4. CREATE FAISS VECTOR DATABASE
```

```
# =====
```

```
print("\nCreating FAISS vector database...")
```

```
dimension = document_embeddings.shape[1]
```

```
index = faiss.IndexFlatL2(dimension)
```

```
index.add(document_embeddings)
```

```
print("FAISS database created.")
print("Documents stored:", index.ntotal)
```

```
# =====
# 5. LOAD LANGUAGE MODEL
# =====
```

```
print("\nLoading language model...")
```

```
llm = pipeline(
    "text2text-generation",
    model="google/flan-t5-small"
)
```

```
print("Language model loaded.")
```

```
# =====
# 6. CHATBOT
# =====
```

```
print("\n=====")
print("      CHATBOT READY")
print("=====")
```

```
print("Ask questions about Artificial Intelligence.")
print("Type 'exit' to stop.")
```

```
while True:
```

```
    question = input("\nYou: ")
```

```
    # Exit chatbot
```

```
    if question.lower() == "exit":
```

```
        print("\nChatbot: Goodbye!")
```

```
        break
```

```
    # =====
```

```
    # 7. CREATE QUERY EMBEDDING
```

```
    # =====
```

```
    query_embedding = embedding_model.encode(  
        [question]  
    )
```

```
    query_embedding = np.array(  
        query_embedding  
    ).astype("float32")
```

```
    # =====
```

```
    # 8. SEARCH FAISS DATABASE
```

```
    # =====
```

```
k = 2
```

```
distances, indices = index.search(  
    query_embedding,  
    k  
)
```

```
# =====  
# 9. GET RELEVANT DOCUMENTS  
# =====
```

```
context = ""
```

```
for i in indices[0]:
```

```
    context = context + documents[i] + "\n"
```

```
# =====  
# 10. CREATE PROMPT  
# =====
```

```
prompt = ""
```

Answer the question using only the context.

Context:

```
{}
```

Question:

}

Answer:

```
"".format(context, question)
```

```
# =====
```

```
# 11. GENERATE ANSWER
```

```
# =====
```

```
result = llm(
```

```
    prompt,
```

```
    max_new_tokens=100
```

```
)
```

```
answer = result[0]["generated_text"]
```

```
# =====
```

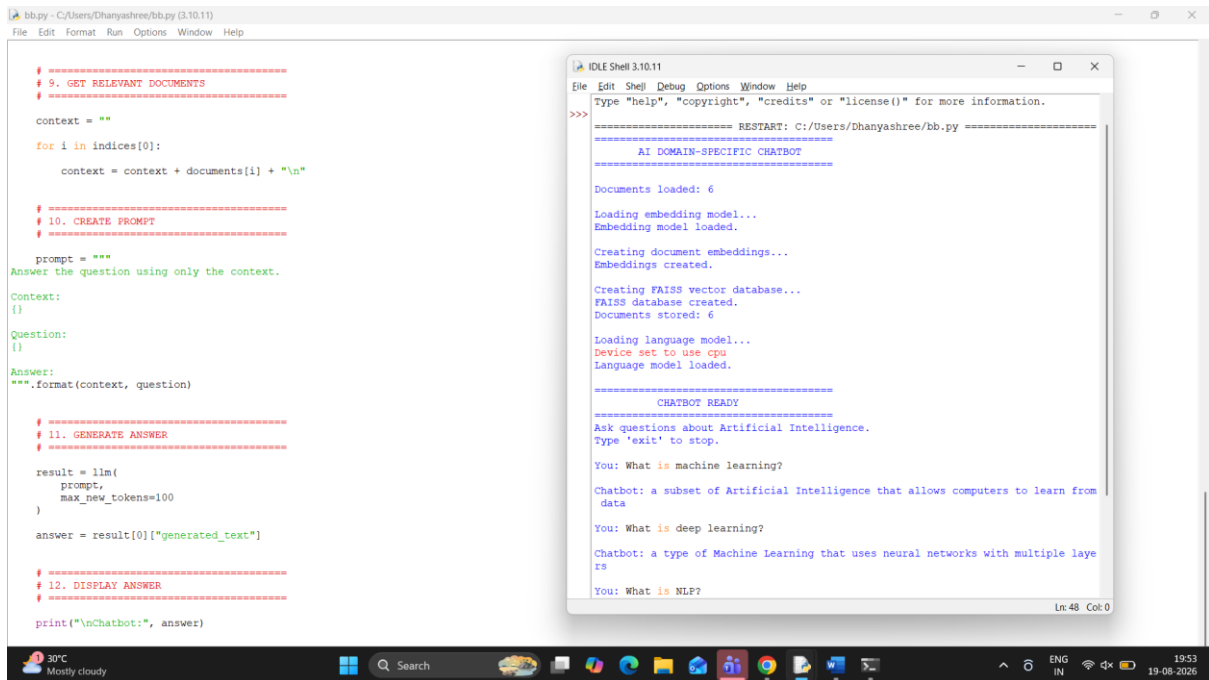
```
# 12. DISPLAY ANSWER
```

```
# =====
```

```
print("\nChatbot:", answer)
```

```
print("\nProgram completed.")
```

output:



The image shows a screenshot of a Windows desktop with two windows open. The background window is IDLE 3.10.11, showing a Python script for a chatbot. The script includes comments for steps 9, 10, 11, and 12, and contains code for loading documents, creating a prompt, and generating an answer. The foreground window is a terminal window titled 'IDLE Shell 3.10.11', showing the output of the script. The output displays the chatbot's initialization process, including loading documents, creating embeddings, and a chatbot ready message. It then shows three sample interactions: 'What is machine learning?', 'What is deep learning?', and 'What is NLP?', each followed by a detailed answer from the chatbot.

```
# =====
# 9. GET RELEVANT DOCUMENTS
# =====

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for i in indices[0]:

    context = context + documents[i] + "\n"

# =====
# 10. CREATE PROMPT
# =====

prompt = """
Answer the question using only the context.
Context:
{}
Question:
{}
Answer:
""".format(context, question)

# =====
# 11. GENERATE ANSWER
# =====

result = llm(
    prompt,
    max_new_tokens=100
)

answer = result[0]["generated_text"]

# =====
# 12. DISPLAY ANSWER
# =====

print("\nChatbot:", answer)
```

```
IDLE Shell 3.10.11
File Edit Shell Debug Options Window Help
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:/Users/Dhanyashree/bb.py =====
AI DOMAIN-SPECIFIC CHATBOT
=====

Documents loaded: 6
Loading embedding model...
Embedding model loaded.
Creating document embeddings...
Embeddings created.
Creating FAISS vector database...
FAISS database created.
Documents stored: 6
Loading language model...
Device set to use cpu
Language model loaded.

=====
CHATBOT READY
=====
Ask questions about Artificial Intelligence.
Type 'exit' to stop.

You: What is machine learning?
Chatbot: a subset of Artificial Intelligence that allows computers to learn from data

You: What is deep learning?
Chatbot: a type of Machine Learning that uses neural networks with multiple layers

You: What is NLP?
Ln: 48 Col: 0
```